

Sharp Fourier sampling numbers in the full linear regime

Candidate partial result for arXiv:2508.13991

9 August 2026

Status. `candidate_partial_likely_valid`. The source's full open problem is not solved. We determine the exact asymptotic rate throughout the parameter sector $q \geq 2$ or $p \leq q$, under only the sharp compact-embedding condition.

1 Source problem and theorem

Equip $\mathbb{T}^d = (\mathbb{R}/2\pi\mathbb{Z})^d$ with normalized measure. Let

$$K_q^s = \{f : \|f\|_{B_\infty^s(L_q)} \leq 1\}$$

be the unit ball of the periodic Besov space, and define its Fourier sampling numbers in L_p by

$$s_n^F(K_q^s)_{L_p} = \inf_{\xi_1, \dots, \xi_n \in \mathbb{Z}^d} \sup \left\{ \|f\|_{L_p} : f \in K_q^s, \widehat{f}(\xi_j) = 0 \ (1 \leq j \leq n) \right\}. \quad (1)$$

Siegel determines nearly sharp bounds under $s/d > 1 - 1/p$ and explicitly leaves the remaining smoothness range open.

The goal of this work is to prove the following bound on the Fourier sampling numbers of Besov balls K_q^s in L_p . As mentioned before, we will restrict ourselves here to the setting where the Gelfand widths are given by (1.8), i.e., where $s > 1 - 1/p$. Determining the correct asymptotics for the Fourier sampling numbers outside of this regime is left as an open problem.

Theorem 1. *Suppose that $1 \leq q, p \leq \infty$ and $s/d > 1 - 1/p$. Then for $n \geq 4$ we have the following upper bounds on the Fourier sampling numbers:*

$$s_n^F(K_q^s)_{L_p} \leq C \begin{cases} n^{-s/d+(1/q-1/p)_+} & q \geq 2 \text{ or } p \leq q \\ n^{-s/d} \log(n)^{1-1/p} \log(\log(n))^{5(1-1/p)} & 1 \leq q < p \leq 2 \\ n^{-s/d+(1/2-1/p)} \sqrt{\log(n)} \log(\log(n))^{5/2} & 1 \leq q < 2 < p. \end{cases} \quad (1.13)$$

Here the constant in (1.13) depends only upon s, q, p and d , but not on n , and the logarithms are taken to base 2.

Let us remark on the choice of (nearly) optimal frequencies, i.e., optimal up to a constant factor, which attain the upper bound (1.13). In the first regime, where either $q \geq 2$ or $p \leq q$, a nearly optimal choice is given by taking the lowest block of frequencies, i.e., the set

$$\{\xi_i\}_{i=1}^n = \{\xi \in \mathbb{Z}^d : |\xi|_\infty \leq m\}, \quad (1.14)$$

where m is chosen such that $n \geq (2m+1)^d$.

Real crop from page 3 of the source paper.

Theorem 1. *Let $d \in \mathbb{N}$, $s > 0$, and $1 \leq p, q \leq \infty$. Suppose*

$$\frac{s}{d} > \left(\frac{1}{q} - \frac{1}{p} \right)_+ \quad (2)$$

and either $p \leq q$ or $q \geq 2$. Then, for $n \geq 2$,

$$s_n^F(K_q^s)_{L_p} \asymp n^{-s/d+(1/q-1/p)_+}. \quad (3)$$

The constants may depend on d, s, p, q , but not on n .

Thus (3) fills every omitted parameter point in the linear regime for which the Besov ball is compact in L_p . In particular, it covers

$$\left(\frac{1}{q} - \frac{1}{p}\right)_+ < \frac{s}{d} \leq 1 - \frac{1}{p},$$

whenever this strip is nonempty and $q \geq 2$ or $p \leq q$.

Proof intuition. The source's low-frequency upper bound already needs only (2). For the matching lower bound, challenge an arbitrary set of sampled frequencies with a block containing twice as many lattice points. When $p \leq q$, a single unsampled exponential has the required size. When $q \geq 2$ and $p > q$, sum all unsampled exponentials in the block. The sum is coherent near the origin, producing the L_p gain, while interpolation between L_2 and L_∞ controls its L_q norm.

2 Two standard estimates

Let $\mathcal{T}_m^d$ denote the trigonometric polynomials supported in $\{k \in \mathbb{Z}^d : |k|_\infty \leq m\}$. We use the standard approximation characterization

$$\|f\|_{B_\infty^s(L_q)} \asymp \|f\|_{L_q} + \sup_{m \geq 0} (m+1)^s E_m(f)_q, \quad (4)$$

where $E_m(f)_q = \inf_{P \in \mathcal{T}_m^d} \|f - P\|_{L_q}$. We also use the de la Vallée Poussin operator V_m : it is supported on $[-m, m]^d$, is uniformly bounded on every L_r , and satisfies

$$\|f - V_m f\|_{L_r} \lesssim E_{m/2}(f)_r. \quad (5)$$

We record explicitly the inverse estimate needed for the witnesses.

Lemma 1. *If $P \in \mathcal{T}_{2N}^d$, $N \geq 1$, then*

$$\|P\|_{B_\infty^s(L_q)} \leq C_{d,s,q} N^s \|P\|_{L_q}. \quad (6)$$

Proof. Choose an integer $r > s$. Bernstein's inequality for trigonometric polynomials and the trivial difference estimate give

$$\omega_r(P, t)_q \lesssim \min\{1, (Nt)^r\} \|P\|_{L_q}.$$

Consequently

$$\sup_{t>0} t^{-s} \omega_r(P, t)_q \lesssim N^s \|P\|_{L_q}.$$

Adding the L_q term proves (6). The argument includes $q = 1, \infty$. □

3 Proof of Theorem 1

Upper bound. First suppose $p \leq q$. Choose the largest even m for which $(2m+1)^d \leq n$, sample every frequency in $[-m, m]^d$, and add arbitrary frequencies if needed. If $f \in K_q^s$ vanishes on the sampled coefficients, then $V_m f = 0$. By normalized-measure Hölder, (5), and (4),

$$\|f\|_{L_p} \leq \|f\|_{L_q} = \|f - V_m f\|_{L_q} \lesssim m^{-s} \lesssim n^{-s/d}. \quad (7)$$

Now suppose $q \geq 2$ and $p > q$. Put

$$t = s - d \left(\frac{1}{q} - \frac{1}{p} \right) > 0.$$

The standard Besov embedding

$$B_\infty^s(L_q) \hookrightarrow B_\infty^t(L_p) \quad (8)$$

and the same low-frequency argument, now in L_p , give

$$\|f\|_{L_p} \lesssim m^{-t} \lesssim n^{-s/d+1/q-1/p}. \quad (9)$$

This proves the upper half of (3). Notice that (2), rather than the source theorem's stronger assumption, is exactly what makes $t > 0$.

Lower bound: construction of an unsampled block. Fix any proposed set $\Lambda \subset \mathbb{Z}^d$ of at most n sampled frequencies. Let N be the least positive integer with $N^d \geq 2n$ and set

$$A_N = \{k \in \mathbb{Z}^d : N \leq k_1 < 2N, \quad 0 \leq k_j < N \ (2 \leq j \leq d)\}. \quad (10)$$

Then $|A_N| = N^d$, $N \asymp n^{1/d}$, and

$$S = A_N \setminus \Lambda, \quad M = |S| \geq \frac{1}{2}N^d. \quad (11)$$

If $p \leq q$, choose $k \in S$. The polynomial $e^{ik \cdot x}$ has every L_r -norm equal to one, and Lemma 1 gives Besov norm at most CN^s . Hence a constant multiple of $N^{-s}e^{ik \cdot x}$ belongs to K_q^s , vanishes on Λ , and has L_p -norm comparable to

$$N^{-s} \asymp n^{-s/d}. \quad (12)$$

It remains to treat $q \geq 2$ and $p > q$. Define

$$D_S(x) = \sum_{k \in S} e^{ik \cdot x}. \quad (13)$$

Orthogonality and the pointwise triangle inequality imply

$$\|D_S\|_{L_2} = M^{1/2}, \quad \|D_S\|_{L_\infty} = M.$$

Interpolation therefore gives

$$\|D_S\|_{L_q} \leq M^{1-1/q}. \quad (14)$$

For the reverse L_p estimate, choose $a = [4(d+1)]^{-1}$. If $|x_j| \leq a/N$ for every j , then for each $k \in A_N$,

$$|k \cdot x| \leq (d+1)a = \frac{1}{4}.$$

Thus all summands in (13) have real part at least $\cos(1/4)$, and

$$|D_S(x)| \geq \cos(1/4)M \quad \text{on a set of normalized measure } c_d N^{-d}.$$

For $p < \infty$ this yields

$$\|D_S\|_{L_p} \gtrsim M N^{-d/p}, \quad (15)$$

and for $p = \infty$ the same formula holds with $N^{-d/p} = 1$, since $D_S(0) = M$. Combining (11), (14), and (15),

$$\frac{\|D_S\|_{L_p}}{\|D_S\|_{L_q}} \gtrsim M^{1/q} N^{-d/p} \gtrsim N^{d(1/q-1/p)}. \quad (16)$$

Lemma 1 shows that

$$f = \frac{D_S}{C N^s \|D_S\|_{L_q}} \quad (17)$$

belongs to K_q^s for a fixed C . Its Fourier coefficients vanish on Λ , while (16) gives

$$\|f\|_{L_p} \gtrsim N^{-s+d(1/q-1/p)} \asymp n^{-s/d+1/q-1/p}. \quad (18)$$

Because Λ was arbitrary, (12) and (18) prove the lower half of (3), completing the proof.

4 Verification, scope, and novelty

Internal checks. The witness is supported in $\mathcal{T}_{2N}^d$, its sampled coefficients vanish identically, and the exponent in (18) is positive in the denominator precisely under (2). A supporting script tested randomly thinned one-dimensional blocks at four scales. Across forty trials per scale, the minimum ratios after division by the predicted powers were 0.633084 for $(q, p) = (2, 4)$, 0.857909 for $(3, 6)$, and 0.707107 for $(2, \infty)$; all removed coefficients were exactly zero. These numerical checks are not used in the proof.

Limitations. The nonlinear sector $q < 2$ and $p > q$ remains open outside the source range. We do not settle the logarithmic factors in that sector, the nonperiodic-domain problem, or the Radon-measurement analogue.

Novelty check and human review. Exact-title and close-variant searches through 9 August 2026 found the source paper but no later statement of (3) under the full compactness condition. The upper estimate is already latent in the source proof; the contribution is the uniform lower-bound witness closing the asymptotics. Human review should check the endpoint inverse estimate in Lemma 1 and search for the dense-unsampled-block argument in older width literature. Novelty is therefore provisional, while the mathematical confidence is high.

References

- [1] J. W. Siegel, *Nearly optimal bounds on the Fourier sampling numbers of Besov spaces*, arXiv:2508.13991v2 (2025).
- [2] R. A. DeVore and G. G. Lorentz, *Constructive Approximation*, Springer, 1993.